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Fourier and applications

1. REMOVE NOISE TO LENA

1.1. Introduction

In this exercise, we are asked to remove the noise from the image "lena_noisy.png" (Figure 1) and make it look like the original photo "lena.png" (Figure 2) after applying this transformation.

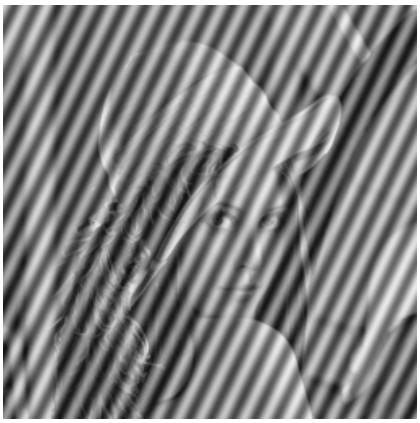

Figure 1: Image of Lena with the noise

Figure 2: Image of Lena with the noise

1.2. First intuition

Our first thought was to match the noise seen in the Lena image with an example explained in the theory (Figure 3). The noise from the theory class had a similar shape, appearing as a black and white wave. The theory included a plot of this noise in the frequency domain (Figure 4), which could simply be described as two dots specifying the frequency values of the noise. So, our first hypothesis was that the aim of the exercise was to find the dots in the frequency domain of Figure 1 and then apply a filter to remove them, hoping this would achieve the goal of the exercise.

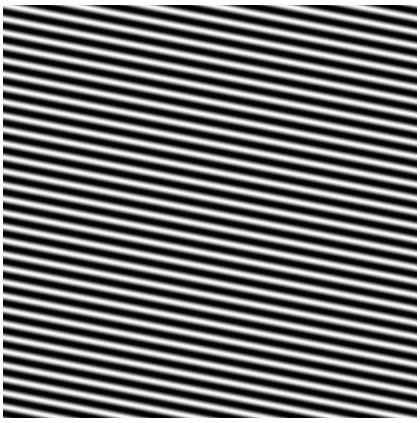

Figure 3: Example of noise explained in the theories

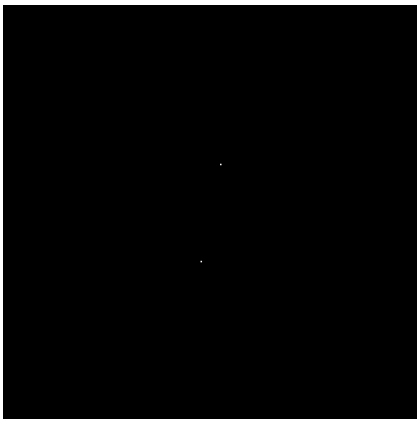

Figure 4: Frequency domain of Figure 3

1.3. First look at our image

To confirm that we were right we applied a FFT (Fast Fourier Transform) to Figure 1 (Figure 5) specting that the result would have the same frequency domain as Figure 1 (Figure 6) but with two dots representing the noise.

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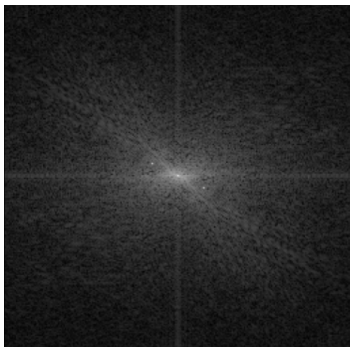

Figure 5: FFT of the noisy Lena image

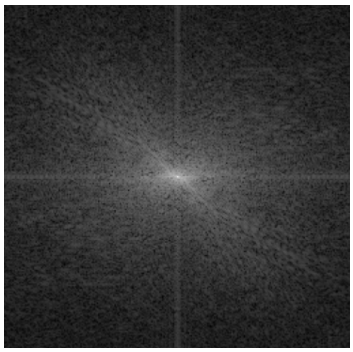

Figure 6: FFT of the original Lena image

As we can see, the result is exactly as we expected. Now thw next step would be remove this two dots from the frequency domain of Figure 1 and then apply an inverse FFT to get the resulting image.

1.4. First intuition of the filter

Taking into account where were places the two dots of Figure 5 we thought that the proper filter explained in the theory should be the BPF (Band Pass Filter). The dots were representing a low frequency respect to some other that could be seen in the original image, but they were not even close the lowest ones. So applying a combination of filters that remove lower or equal frequencies as the frequencies of the dots was a bad idea, because we would be removing a lot of frequencies that are not noise. And the image could be too blurry. On the other hand, applying a BPF that only remove frequencies around the ones of the dots would be a good idea, because we would be removing only the noise and not the rest of the frequencies.

1.5. Creation of the filter

To create the filter was really helpful to learn from the filter examples in the Colab NoteBook. After analyzing some of them we decided that our filter should the sum of a LPF and HPF. In this way, the plot result of the filter should be at most white, but it should have a black circle in the middle representing the frequencies that we want to remove. The rest of the frequencies should be white, because we want to keep them. With the example filters we did a first plot representing the first idea we had for the filter (Figure 7).

We had a lot of trubble to find the right values for the radius of the circles of the LPF and HPF, but after some trial and error we found that a radius of 20 pixels for the LPF and 21.3 pixels for the HPF were strictly the right values to remove the noise without removing too much frequencies. The resulting filter can be seen in Figure 8.

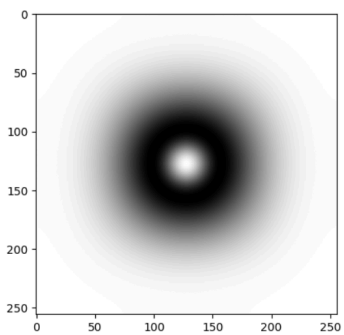

Figure 7: First idea of the filter

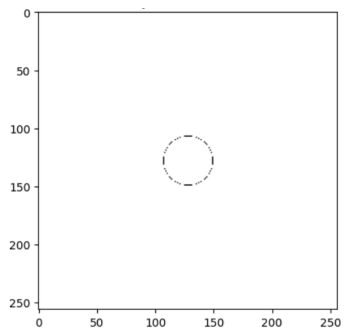

Figure 8: Resulting filter

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1.6. Application of the filter

After creating the filter, we applied it to the FFT of Figure 1 and then we applied an inverse FFT to get the resulting image. The resulting image can be seen in Figure 9. As we can see, the resulting image is really similar to the original Lena image, but it is not exactly the same. We can see that there is still some noise in the resulting image, but it is really reduced compared to the original noisy image. We think that the reason why there is still some noise in the resulting image is because the combination of the two filters is not perfect, due to its circular shape. The ideal filter should be two circles with the same radius as the dots and centered in the same position as the dots. But with the resources given in the theoretical and practical classes, we consider we were not able to create a filter with this shape, so we had to settle for a combination of two filters that was not perfect, but it was good enough to remove most of the noise.

Figure 9: Resulting image after applying the filter

1.7. Extra: Asking AI for the perfect filter

To understand well how our ideal filter would be implemented, we asked to an AI to create a filter that would remove the noise from the image. AI talked about a Notch Filter which was not in the theory classes but had the exactly shape we thought it would have. The resulting filter can be seen in Figure 10. It is interesting to see the plot of the filter applied in the frequency domain of the noisy image (Figure 11). And finally we can observe that the final result it is almost perfect and can be seen in Figure 12.

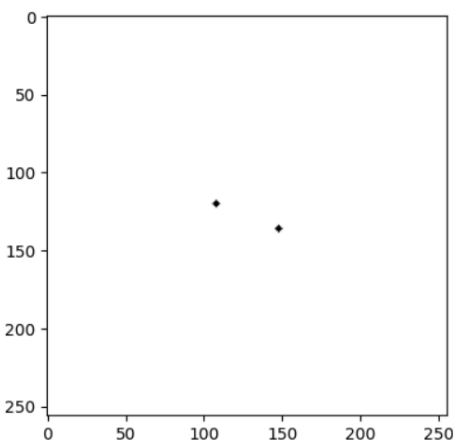

Figure 10: Ideal filter created by AI

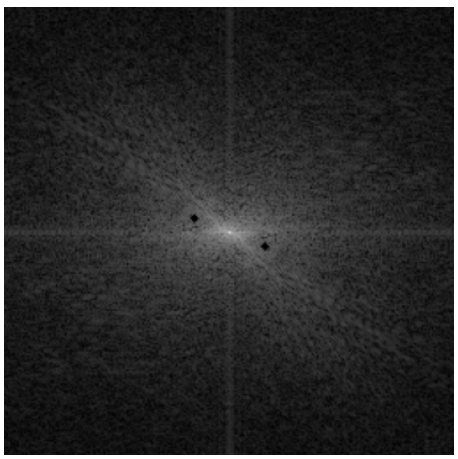

Figure 11: FFT of the noisy image with the ideal filter applied

Figure 12: Resulting image after applying the ideal filter